



APRIL, 2025

REGIONAL MARITIME UNIVERSITY

MARINE ENGINEERING

BSC MARINE AND MECHANICAL ENGINEERING (BME 28B/BMT 28B)

END OF FIRST SEMESTER EXAMINATIONS

WORKSHOP TECHNOLOGY I

INSTRUCTIONS:

TIME ALLOWED: 3HRS

Answer all question in section A and section B.

SECTION A

1. It is dangerous to use chisels that have:
 - A. hard points.
 - B. mushroomed heads.
 - C. small cutting points.
 - D. rough cutting edges.
2. A cold chisel that has a mushroomed head can be reclaimed by:
 - A. regrounding to the recommended included angle.
 - B. grinding to the original shaft contour.
 - C. re-tempering the mushroomed head.
 - D. filing the head to a ball shape.
3. File handles should be on files because:
 - A. they reduce the risk of injury.
 - B. they protect the tang from damage.
 - C. they are available in a variety of attractive colours.
 - D. they protect the cutting edges on the face.
4. A split hammer handle is best repaired by:
 - A. removing and replacing it.
 - B. binding it with nylon string.
 - C. wrapping with copper wire.
 - D. taping it with electrical tape.
5. The safest way to use a wrench is to:
 - A. pull on it with steady pressure.
 - B. push it away from your body.
 - C. tap on it to get it started.
 - D. put a pipe handle on it for extra leverage.

Doc. No.: EXM/F/03	Page No.: 1/7	Issue No.: 1	Last Update: 15/05/15
Revision No.: 2	Date: 12/04/06	Author: EC	Approved: PROVOST

6. A combination square consists of:

- A. a square head.
- B. a protractor.
- C. a centre-head.
- D. all of the above.

7. The most versatile type of square is the:

- A. carpenter's square.
- B. sliding T bevel square.
- C. combination square.
- D. try square.

8. A good tool for accurately transferring measurements is a:

- A. tape measure.
- B. combination square.
- C. vernier caliper.
- D. set of dividers.

9. A flexible steel rule is an advantage when:

- A. measuring long distances.
- B. laying out channel iron.
- C. finding the centres of circles.
- D. measuring on curved surfaces.

10. Vise grips have an advantage over pliers when:

- A. the tool has to grip round material.
- B. a metric size wrench is not available.
- C. you need to maintain a constant pressure.
- D. you are working on non-ferrous material.

11. A bar clamp has an advantage when:

- A. thick metal is being clamped.
- B. clamping over greater distances.
- C. material needs to be clamped to a bar.
- D. pressure is applied in one spot.

12. A 32 teeth per inch hacksaw blade is the best choice for cutting:

- A. hard material.
- B. soft material.
- C. thin stock.
- D. thick stock.

13. Files are classified by the:

- A. job you have been asked to do.
- B. length, shape and cut.
- C. size, width and kind of material.
- D. shape, hardness, thickness and cut.

Doc. No.: EXM/F/03	Page No.: 2/7	Issue No.: 1	Last Update: 15/05/15
Revision No.: 2	Date: 12/04/06	Author: EC	Approved: PROVOST

- are sockets sized?
 by the total number of points
 by measuring the distance across the points
 by measuring the distance across the flats
 D. by the depth of the socket
5. An adjustable wrench is recommended when:
 A. the points of the nut are rounded over.
 B. you are working with metric fasteners.
 C. a proper fitting wrench is not available.
 D. square nuts are used instead of hex nuts.
16. What is the carbon content range of low carbon steels?
 A. 0.05% to 0.30%
 B. 0.30% to 0.45%
 C. 0.45% to 0.75%
 D. 0.75% to 1.50%
17. Punches and dies are made from what class of carbon steel?
 A. low
 B. medium
 C. high
 D. very high
18. What is the carbon content range of high carbon steels?
 A. 0.05% to 0.30%
 B. 0.30% to 0.45%
 C. 0.45% to 0.75%
 D. 0.75% to 1.50%
19. What is the carbon content range of medium carbon steel?
 A. 0.05% to 0.30%
 B. 0.30% to 0.45%
 C. 0.45% to 0.75%
 D. 0.75% to 1.50%
20. Which class of steel is generally tough, ductile, easily formed and usually welded without preheat?
 A. low carbon
 B. medium carbon
 C. high carbon
 D. very high carbon

Doc. No.: EXM/F/03	Page No.: 3/7	Issue No.: 1	Last Update: 15/05/15
Revision No.: 2	Date: 12/04/06	Author: EC	Approved: PROVOST

21. Which elements exist in small quantities in carbon steels and are usually considered as impurities?
- A. iron, carbon, manganese
 - B. silicon, sulphur, manganese
 - C. phosphorus, silicon, manganese
 - D. sulphur, phosphorus
22. Which element in carbon steels, in percentages up to about 0.80%, is intended to combine with Sulphur to offset embrittlement and hot shortness?
- A. phosphorus
 - B. silicon
 - C. manganese
 - D. carbon
23. Which of the following elements is considered as an impurity in high percentages, but slightly improves the strength and corrosion resistance of low carbon steel in low percentages up to approximately 0.01 %?
- A. phosphorus,
 - B. sulphur
 - C. silicon
 - D. manganese
24. The content of this element in carbon steel is usually kept below 0.06%, but a content of 0.1 to 0.15% can improve its machinability.
- A. manganese
 - B. sulphur
 - C. phosphorus
 - D. silicon
25. Which element increases the strength and toughness of steel at low temperatures?
- A. nickel
 - B. chromium
 - C. tungsten
 - D. molybdenum
26. What is the purpose of having a chromium content of 12% to 30% in steel?
- A. increases corrosion resistance
 - B. gives the steel a shiny finish
 - C. increases hardness at high temperatures
 - D. improves toughness at low temperatures
27. Which element is used to decrease high temperature creep (slow stretching of steel under stress at high temperatures)?
- A. cobalt
 - B. manganese
 - C. molybdenum
 - D. silicon

- is the alloying element in carbon steel that tends to produce a fine grain structure during the heat-treating process?
- tungsten
 - titanium
 - chromium
 - vanadium

29. Which element, in percentages of 1 % to 15%, increases the toughness and hardenability of steel?
- magnesium
 - manganese
 - molybdenum
 - mercury

30. Which metric measurement is used to indicate the distance from Edmonton to Calgary?
- millimetres
 - centimetres
 - kilometres
 - decimetres

SECTION B

1. Two holes are to be drilled in a lock template. Calculate dimensions A and B in Figure below.

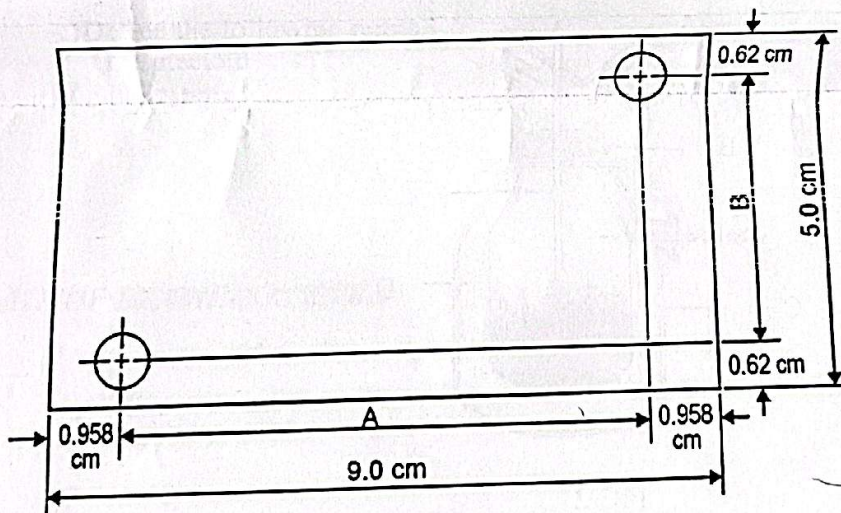


Figure - Lock Template.

Convert the answer to millimeters

Dimension A..... Dimension B.....

Doc. No.: EXM/F/03	Page No.: 5/7	Issue No.: 1	Last Update: 15/05/15
Revision No.: 2	Date: 12/04/06	Author: EC	Approved: PROVOST

2. a) Draw the cubic structures of BCC and FCC, and Explain.
 b) State 4 examples each of materials with BCC and FCC structures.
 c) What is the maximum temperature for BCC to form?
3. a) Explain why filing is done only on the forward stroke.
 b) Label the hand tool in FIG 1 below.

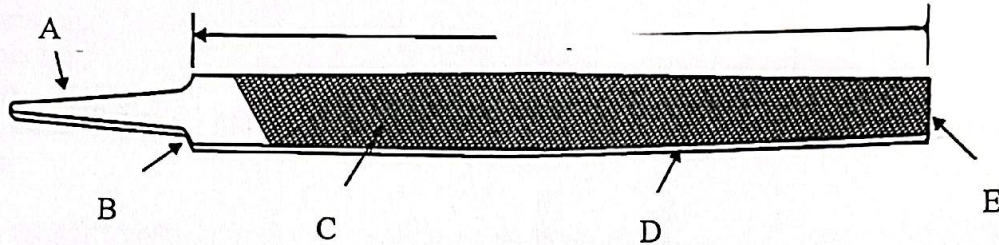


FIG 1

- c) The cutting edge of a cold chisel should be prepared to a degree angle for softer materials and up to degrees for very hard materials.

4. a) Label the parts of the oxygen steel making furnace.

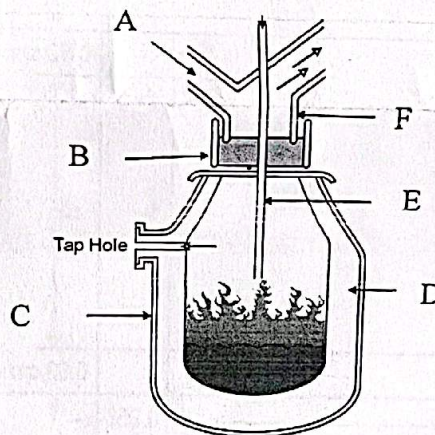


FIG 2

- b) Explain the Basic oxygen steel-making process (BOS).

5. Define the following terms as they relate to the grain structure of steel.
 - i. Ferrite
 - i. Pearlite
 - ii. Cementite
 - iii. Austenite
 - iv. Martensite

Doc. No.: EXM/F/03	Page No.: 6/7	Issue No.: 1	Last Update: 15/05/15
Revision No.: 2	Date: 12/04/06	Author: EC	Approved: PROVOST

a) Complete the phase diagram below by providing all the stages in their right temperature levels.

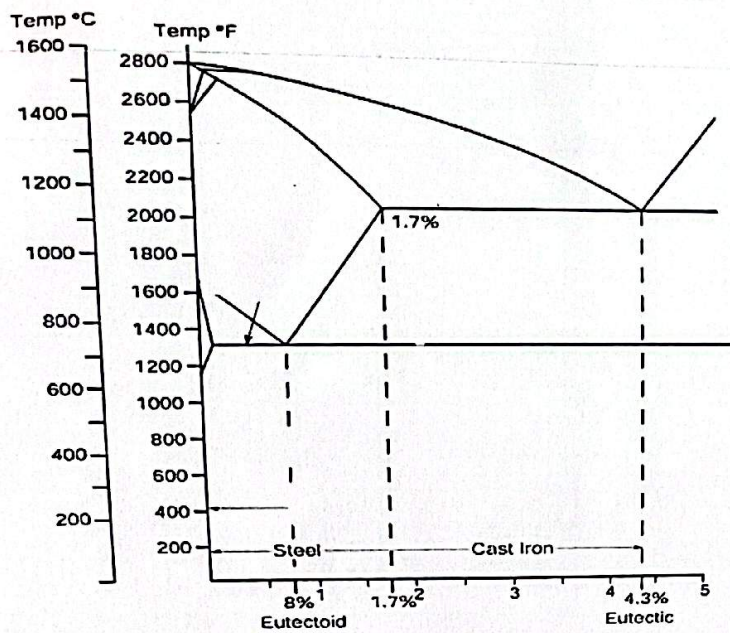


FIG 3

b) Explain the effects of rapid cooling on the grain structure of carbon steel.

c) Define the following terms,

- i. Eutectoid
- ii. Eutectic

INITIALS OF EXAMINER: ENKA

Doc. No.: EXM/F/03	Page No.: 7/7	Issue No.: 1	Last Update: 15/05/15
Revision No.: 2	Date: 12/04/06	Author: EC	Approved: PROVOST